

Efficient Approximation of Extremal Field Maps

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Motivation

- In trajectory optimization, we face the reality that there will always be state uncertainty present.
- Over time, this uncertainty compounds, necessitating a redesign of the trajectory, costing engineers valuable time and effort.
- To avoid having to redesign trajectories on mission, pre-compute a volume of trajectories and map the state to the corresponding optimal solution
- If indirect optimal control methods are leveraged, the optimal solution is characterized by the optimal costate vector and the dynamics, meaning an extremal field map (EFM) can be constructed as a mapping from the states to the costates, simplifying the problem

Methodology

Formulation

- We define an extremal field map (EFM) as a 1-to-1 mapping from a set of states and/or parameters to an optimal solution
- Using **indirect optimal control**, we can represent an optimal solution as an **optimal costate vector** and corresponding equations of motion
- This fact can be used to create an EFM:

$$EFM: \mathbf{P} \rightarrow \lambda^*$$

- For spacecraft navigation, we can define $\mathbf{P}$ to be the state vector, $\mathbf{x}$, and time t , to map the current state to the optimal control sequence which drive the state to the boundary condition.

$$EFM: [\mathbf{x}, t] \rightarrow \lambda^*$$

Construction of Extremal Field Maps

- Extremal Field Maps can be constructed “forwards” or “backwards”
- Forward Construction – at discrete grid points, solve 2-point boundary value problem using a nonlinear program
 - Requires significant compute (hours)
 - Easy to approximate using meshed methods
- Backwards Construction – from a defined boundary condition (typically a rendezvous), perturb the nominal solution and propagate backwards in time to generate a neighboring trajectory
 - Easier to compute (minutes) since it only requires propagation
 - Requires a mesh-free approximation method since data points will not be evenly spaced

→ Here, the reverse construction procedure is used.

Reverse Construction Procedure:

- Define a fixed final boundary condition
- Create N perturbed final costate vectors
 - Use a nominal solution as reference
- Propagate all the perturbed final state/costate vectors backwards in time
- Interpolate the data generated from propagation using a mesh-free approximation method

Creating a Continuous Approximation

- For reverse construction, we need a **mesh-free approximation method**.
- Due to the unpredictability of costate surfaces, choose **polyharmonic spline interpolation** to obtain high accuracy at multiple resolutions
 - Polyharmonic splines are a subset of radial basis functions. They combine a **local radial basis function** kernel with a **global polynomial term** to achieve high accuracy with no need for a mesh

$$\mathbf{y} \approx \Phi \boldsymbol{\alpha} + B \mathbf{v}$$

$$\Phi = \begin{bmatrix} \phi(\|\mathbf{x}_1 - \mathbf{x}_1\|) & \phi(\|\mathbf{x}_1 - \mathbf{x}_2\|) & \dots & \phi(\|\mathbf{x}_1 - \mathbf{x}_N\|) \\ \phi(\|\mathbf{x}_2 - \mathbf{x}_1\|) & \phi(\|\mathbf{x}_2 - \mathbf{x}_2\|) & \dots & \phi(\|\mathbf{x}_2 - \mathbf{x}_N\|) \\ \vdots & \vdots & \ddots & \vdots \\ \phi(\|\mathbf{x}_N - \mathbf{x}_1\|) & \phi(\|\mathbf{x}_N - \mathbf{x}_2\|) & \dots & \phi(\|\mathbf{x}_N - \mathbf{x}_N\|) \end{bmatrix} B = \begin{bmatrix} 1 & \mathbf{x} & \mathbf{x}^2 & \dots \\ 1 & \mathbf{x} & \mathbf{x}^2 & \dots \\ 1 & \mathbf{x} & \mathbf{x}^2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Where $\boldsymbol{\alpha}$ and $\mathbf{v}$ are weight vectors, fit with least squares:

$$\begin{bmatrix} \Phi & B \\ B^T & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\alpha} \\ \mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{y} \\ \mathbf{0} \end{bmatrix}$$

The second row of this block linear system enforces a constraint that $\mathbf{v}$ must be in the null space of B^T . Conveniently, we can fold this constraint into the top row to simplify the system and force the resulting system ($Z^T \Phi Z$) to be **symmetric positive definite**:

$$\Phi \boldsymbol{\alpha} + B \mathbf{v} = \Phi Z \boldsymbol{\gamma} + B \mathbf{v} = \mathbf{y} \quad \text{where} \quad B^T Z = \mathbf{0}$$

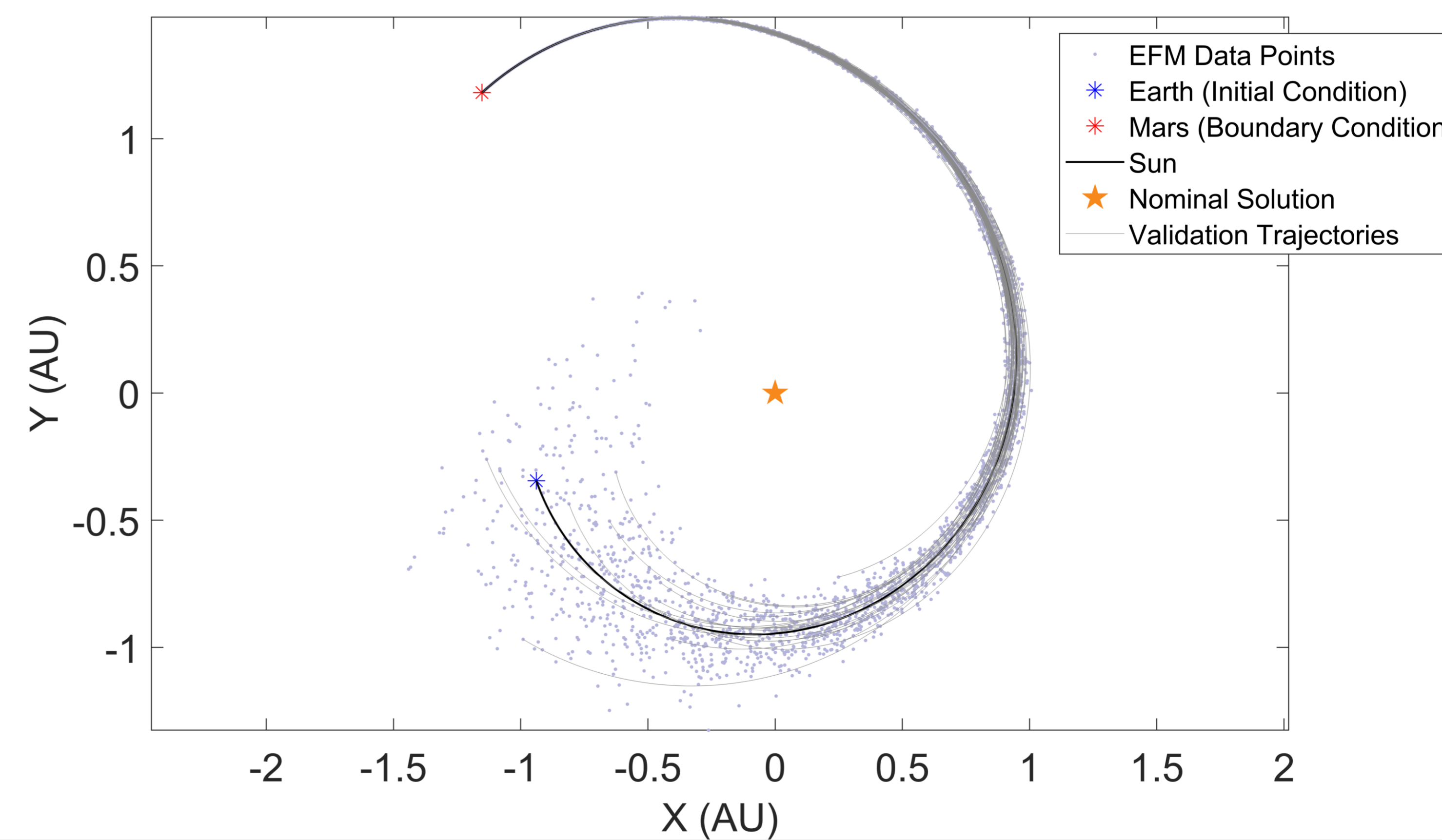
$$Z^T \Phi Z \boldsymbol{\gamma} = Z^T \mathbf{y}$$

Then solve this system for the original weight vectors

$$\boldsymbol{\alpha} = Z \boldsymbol{\gamma} \quad B \mathbf{v} = \mathbf{f} - \Phi Z \boldsymbol{\gamma}$$

This method requires an additional matrix inverse, but is still much faster due to the symmetric positive definite property

Results

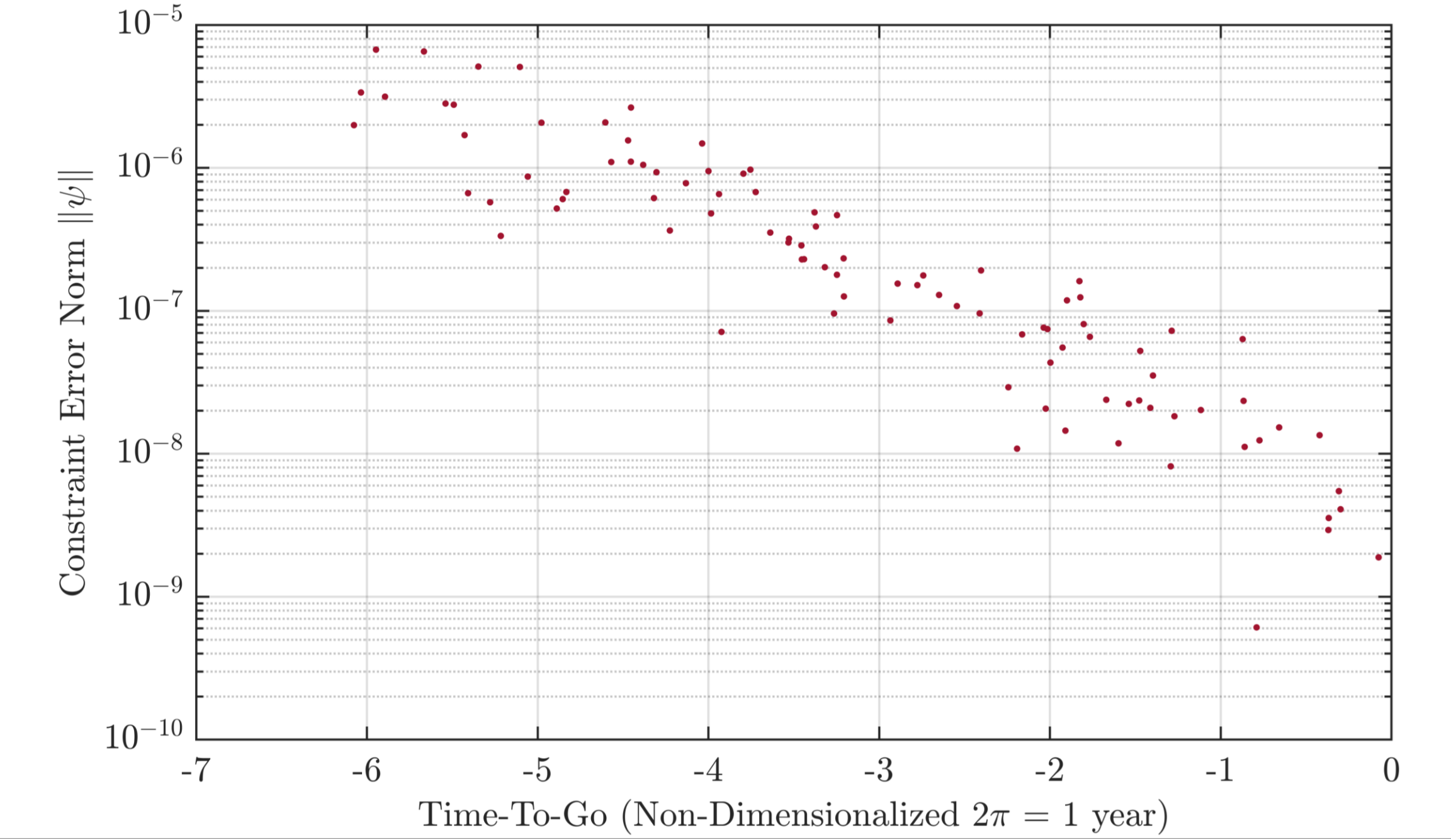


Using reverse construction and the polyharmonic spline interpolation method for the Earth-to-Mars minimum energy problem (4000 data points, 2nd order polynomial term) we obtain a continuous volume of optimal costate approximation; which, if evaluated using 100 test initial states within the volume, achieves a final constraint error of:

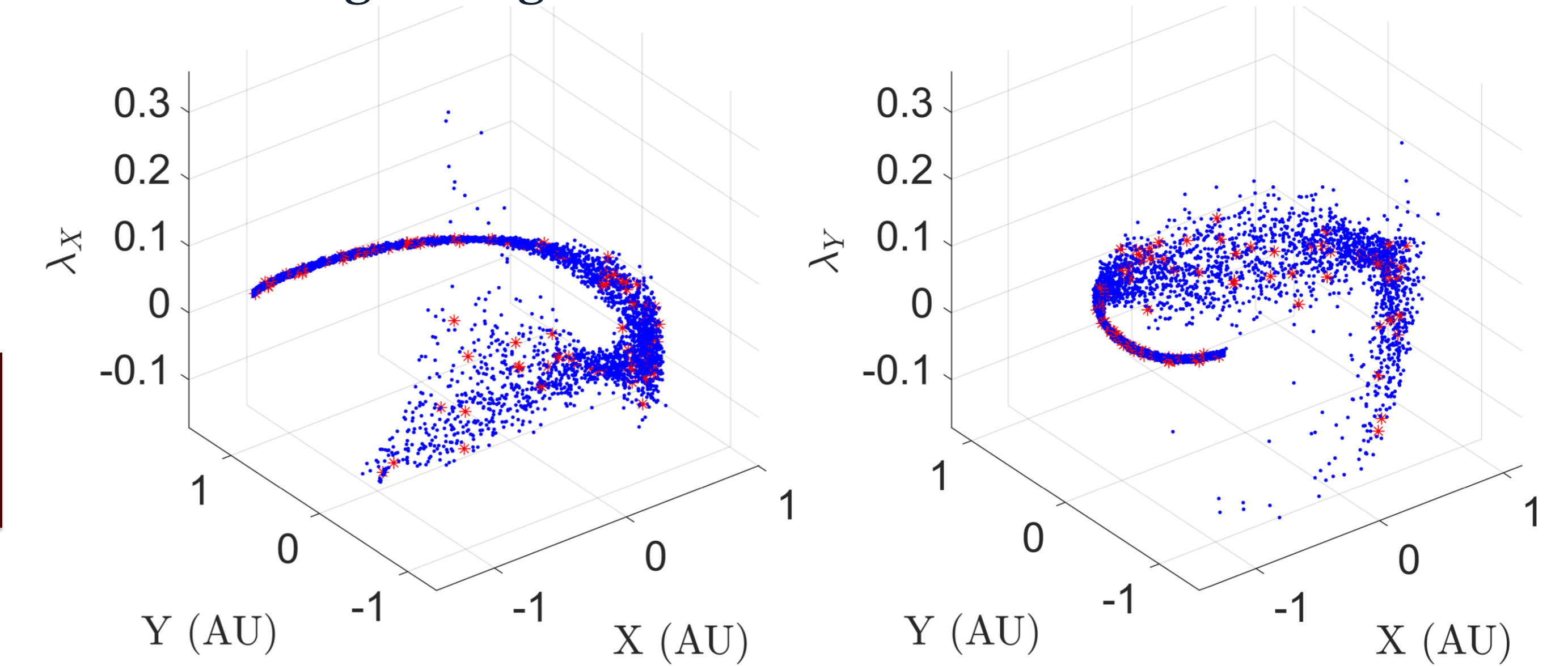
$$\|\boldsymbol{\psi}\| = \begin{bmatrix} \mathbf{r}(t_f) - \mathbf{r}_f \\ \mathbf{v}(t_f) - \mathbf{v}_f \end{bmatrix} = 1.7 \times 10^{-5}$$

$$\boldsymbol{\psi}_{\text{mean absolute}} = 3.7 \times 10^{-7}$$

We can also observe that, predictably, the **constraint error is linearly related to time-of-flight**. This means we could reduce final error greatly by re-sampling the extremal field map repeatedly during the mission.



We can also directly observe the geometry of the costate surfaces to get insight into the state sensitivities



Conclusions

- The **reverse construction** method **drastically reduces compute time** when compared to forward construction
- Using reverse construction demands a mesh-free approximation method, polyharmonic spline interpolation achieves **high accuracy**
- EFM **accuracy decreases with time-of-flight**, meaning an EFM guidance system would benefit heavily from in-the-loop resampling
- Accurate EFMs can be created with only a few thousand points and a single layer approximation.

Key References

- [1] S. K. Singh and J. L. Junkins, “Stochastic learning and extremal-field map based autonomous guidance of low-thrust spacecraft,” Scientific Reports, Vol. 12, 10 2022
- [2] A. Widmer, T. L. Ahrens, and J. L. Junkins, “Orthogonal Approximation of Extremal Field Maps,” IAA Astrodynamics Specialist Conference, ASC 2024, Univert Inc., American Astronautical Society.
- [4] D. Izzo and E. Ozturk, “Real-Time Guidance for Low-Thrust Transfers Using Deep Neural Networks,” Journal of Guidance, Control, and Dynamics, Vol. 44, No. 2, 2021, pp. 315–327, 10.2514/1.6005254.
- [9] M. Powell, “Some algorithms for thin plate spline interpolation to functions of two variables,” Advances in Computational Mathematics, New Delhi, India (ed. HP Dikshit & CA Micchelli), 1994, pp. 303–319
- [12] S. Lloyd, “Least squares quantization in PCM,” IEEE Transactions on Information Theory, Vol. 28, No. 2, 1982, pp. 129–137, 10.1109/TIT.1982.1056489